ACADEMIC CENTRAL

PRACTICE EXAMINATION

UNIVERSITY · 1ST YEAR

DIGITAL ELECTRONICS

PAPER 1 · PRACTICE A · MARKING GUIDELINE

MARKS: 100

Marking guideline for the Academic Central University · 1st year Digital Electronics paper 1 · practice a.

Written by Academic Central for practice, following first-year Digital Electronics (digital systems) syllabuses at South African universities (BEng Electrical, Electronic and Mechatronic Engineering). It is not a university memorandum.

How to mark: each ✓ is one mark. Give marks for correct steps even if a later step is wrong (consistent accuracy). Accept any correct method, equivalent expressions and equivalent circuits.

QUESTION 1

1.1 156 = 128 + 16 + 8 + 4 ✓ = $1001\ 1100_2$ ✓ = $9C_{16}$ ✓ (3)

1.2 128 + 32 + 16 + 4 + 2 ✓ = 182 ✓ (2)

1.3 3 × 256 + 10 × 16 + 15 ✓ = 943 ✓ (2)

1.4 +45 = 0010 1101 ✓ invert: 1101 0010 ✓ add 1: 1101 0011 ✓ (3)

1.5 0110 1011 + 0011 0101 = 1010 0000 ✓ (2)
Check: 107 + 53 = 160 = 1010 0000 ✓

1.6 0100 0111 ✓ (1)

1.7 Only ONE bit changes between neighbouring positions, so a reading taken while the shaft is moving cannot jump to a wrong value ✓ (2)
0111 → 0100 ✓

[15]

QUESTION 2

2.1 (3)

A	B	NAND	NOR	XOR
0	0	1	1	0
0	1	1	0	1
1	0	1	0	1
1	1	0	0	0

NAND ✓ NOR ✓ XOR ✓

2.2 Any logic function can be built from NAND gates only ✓ (3)
NOT: join both inputs of a NAND gate together ✓
AND: a NAND gate followed by a NAND gate wired as a NOT ✓

2.3 $Y = A(B + B') + A'B$ distributive law ✓ (4)
$= A \cdot 1 + A'B = A + A'B$ complement and identity laws ✓
$= (A + A')(A + B)$ distributive law ✓
$= A + B$ ✓

2.4 (3)

A	B	A + B	(A + B)′	A′	B′	A′ · B′
0	0	0	1	1	1	1
0	1	1	0	1	0	0
1	0	1	0	0	1	0
1	1	1	0	0	0	0

(A + B)′ column ✓ A′ · B′ column ✓ the two columns are identical, so the theorem holds ✓

2.5.1 X = A · B + B′ · C ✓✓ (2)

2.5.2 (4)

A	B	C	X
0	0	0	0
0	0	1	1
0	1	0	0
0	1	1	0
1	0	0	0
1	0	1	1
1	1	0	1
1	1	1	1

One mark for every two correct rows ✓✓✓✓

2.5.3 X = 1 ✓ (1)

[20]

QUESTION 3

3.1 (6)

AB \ CD	00	01	11	10
00	1	0	0	1
01	0	1	1	0
11	0	1	1	0
10	1	0	0	1

Map filled in correctly ✓✓
The four corners (m0, m2, m8, m10) make a group of 4: B′D′ ✓✓
The middle square (m5, m7, m13, m15) makes a group of 4: BD ✓
F = B′D′ + BD (this is B XNOR D) ✓

3.2.1 (3)

A	B	C	Y
0	0	0	0
0	0	1	0
0	1	0	0
0	1	1	1
1	0	0	0
1	0	1	1
1	1	0	1
1	1	1	1

Inputs ✓ output column ✓✓

3.2.2 Y = A′BC + AB′C + ABC′ + ABC ✓✓ (2)

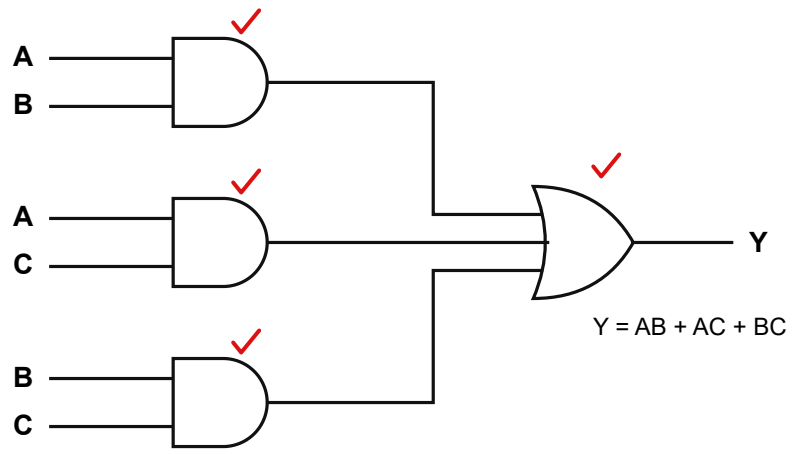
3.2.3 ABC is used three times (X + X = X): (3)

Y = (A′BC + ABC) + (AB′C + ABC) + (ABC′ + ABC) ✓

= BC(A′ + A) + AC(B′ + B) + AB(C′ + C) ✓

= AB + AC + BC ✓ (a K-map gives the same answer)

3.2.4



Three 2-input AND gates (AB, AC, BC) ✓✓; a 3-input OR gate combining them ✓

(3)

3.2.5 Write Y as a double complement: Y = ((AB)′ · (AC)′ · (BC)′)′ by De Morgan's theorem ✓ (3)

Replace the three AND gates with 2-input NAND gates ✓

Replace the OR gate with a 3-input NAND gate fed by their outputs (NAND–NAND form) ✓

[20]

QUESTION 4

4.1 (4)

A	B	S	C
0	0	0	0
0	1	1	0
1	0	1	0
1	1	0	1

Truth table ✓✓ $S = A \oplus B$ ✓ $C = A \cdot B$ ✓

4.2 The first half adder adds A and B; the second adds its sum to C_{in}, giving the final sum ✓. The two carry outputs are joined by an OR gate to give C_{out} ✓ (3)
$1 + 1 + 1 = 11_2$, so S = 1 and $C_{out} = 1$ ✓

4.3.1 $Y = S_1'S_0'I_0 + S_1'S_0I_1 + S_1S_0'I_2 + S_1S_0I_3$ ✓✓✓ (3)

4.3.2 I_2 ✓ (1)

4.4 Any ONE: address decoding (selecting one of four memory or input/output chips); selecting one of four devices or displays from two output pins; driving one of four outputs ✓✓ (2)

4.5 a, b and c ✓✓ (2)

[15]

QUESTION 5

5.1 In combinational logic, the output depends only on the present inputs ✓. In sequential logic, the output depends on the present inputs AND on the previous state (it has memory) ✓ (2)

5.2 (4)

S	R	Q	Action
0	0	Q (no change)	hold
0	1	0	reset
1	0	1	set
1	1	–	not allowed

Truth table ✓✓
S = R = 1 is not allowed ✓, because both outputs are forced to 0 (Q and Q′ are no longer complements) and the final state is unpredictable when both inputs return to 0 ✓

5.3 Q: 1, 0, 1, 1, 0 ✓✓ (3)
Edge-triggered: the flip-flop only copies D to Q at the moment the clock changes from 0 to 1, not while the clock is high ✓

5.4 (4)

J	K	Q after the clock edge
0	0	Q (no change)
0	1	0 (reset)
1	0	1 (set)
1	1	Q′ (toggle)

Truth table ✓✓✓
Toggle mode: with J = K = 1 the output changes to the opposite state on every clock edge, so it divides the clock frequency by 2 ✓

5.5.1 $2^3 = 8$ states ✓ highest count 7 (111) ✓ (2)

5.5.2 Detect the count 110 (6) with a gate and use it to reset (clear) all the flip-flops, so it counts 0 to 5 ✓ (1)

5.6 f = 1 MHz / 2^4 ✓ = 62.5 kHz ✓ (2)

5.7 $Q_3Q_2Q_1Q_0$ = 1011 ✓✓ (2)

[20]

QUESTION 6

6.1 Supply: TTL needs 5 V ± 0.25 V; CMOS works over a wide range, e.g. 3 V to 15 V ✓ (4)
Power: CMOS uses much less power, especially when it is not switching ✓; TTL uses more ✓
Static: CMOS is easily damaged by static electricity; TTL is less sensitive ✓

6.2 The largest number of standard gate inputs that one gate output can drive ✓ while still keeping correct logic levels ✓ (2)

6.3 R = (5 − 2) / 0.01 = 300 Ω ✓ use the next standard value up, 330 Ω ✓ (2)

6.4 It holds the input at a definite logic 1 when the button is not pressed ✓, so the input does not "float" and read random values ✓ (2)

[10]

TOTAL: 100

END OF MARKING GUIDELINE